

Problem 2.2

It is known that $a(t)$ is of the form

$$a(t) = at^2 + b.$$

If \$100 invested at time 0 accumulates to \$172 at time 3, find the accumulated value at time 10 of \$100 invested at time 5.

Solution

Let $A(t)$ denote the accumulated value at time t , and let $a(t)$ be the accumulation function. Then

$$A(t) = A(0)a(t).$$

Also, by definition of the accumulation function,

$$a(0) = 1.$$

Since

$$a(t) = at^2 + b,$$

we substitute $t = 0$ to get

$$a(0) = a(0)^2 + b = b.$$

Hence,

$$b = 1.$$

Therefore,

$$a(t) = at^2 + 1.$$

Now we use the information that \$100 invested at time 0 accumulates to \$172 at time 3:

$$172 = A(3) = 100 a(3).$$

So,

$$1.72 = a(3) = a(3)^2 + 1 = 9a + 1.$$

Thus,

$$1.72 - 1 = 9a,$$

which gives

$$0.72 = 9a, \quad a = 0.08.$$

Hence the accumulation function is

$$\boxed{a(t) = 0.08t^2 + 1}.$$

Now compute:

$$a(5) = 0.08(25) + 1 = 3,$$

and

$$a(10) = 0.08(100) + 1 = 9.$$

An amount of \$100 invested at time 5 grows from time 5 to time 10 by the factor

$$\frac{a(10)}{a(5)}.$$

Therefore, the accumulated value at time 10 is

$$100 \cdot \frac{a(10)}{a(5)} = 100 \left(\frac{9}{3} \right) = 300.$$

Hence, the required accumulated value is

$$\boxed{\$300}.$$